Do Critics and Audiences Agree?

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1 Introduction

For this project I used a Rotten Tomatoes dataset from Kaggle. I chose it because, every time I watch a movie with friends, we end up discussing whether it was actually good, and we often disagree with what the critics said. This made me wonder whether this is common. Rotten Tomatoes is a website that provides scores for movies and TV shows. It gives two scores between 0 and 100: a critic score (the Tomatometer), based on professional reviews, and an audience score, based on ratings from viewers. This project asks whether critics and audiences agree, using two subquestions:

1. Do critics and audiences give different average scores?
2. Does this depend on the genre of the movie?

2 Data and methods

The dataset `rotten_tomatoes_movies.csv` contains 17,712 movies and 22 columns. I selected three of them: `tomatometer_rating`, `audience_rating` and `genres`. Movies without both scores were removed and values outside 0 and 100 were removed, leaving 17,407 movies (98.3%). Each movie was given one genre, the first one listed. For every movie the difference between critic score and audience score was calculated (d). A positive d means critics liked the movie more. For a group of n movies I then calculated:

- The mean difference $\bar{d}$;
- Its uncertainty, the standard error of the mean, SEM $= s/\sqrt{n}$, where s is the standard deviation;
- The significance, $\bar{d}$/SEM, in units of σ.

A difference of 3σ or more considered as a real difference. The same calculation was repeated for each genre with at least 30 movies. As a stability check, the movies were split into 10 random subsamples of about 1,740 movies. All analysis and plots were conducted with Python (`rotten_tomatoes.py`).

3 Results

Do critics and audience give different average scores? The mean difference is 0.19 ± 0.16 points, only 1.2σ from zero (Figure 1). So on average critics and audiences give the same scores. Individual movies, however, differ a lot: the standard deviation is 21.6 points. The differences go in both directions and cancel out in the average. The 10 random subsamples all gave means close to zero (between -0.69 and 1.08, none above 2.1σ), so the result is stable.

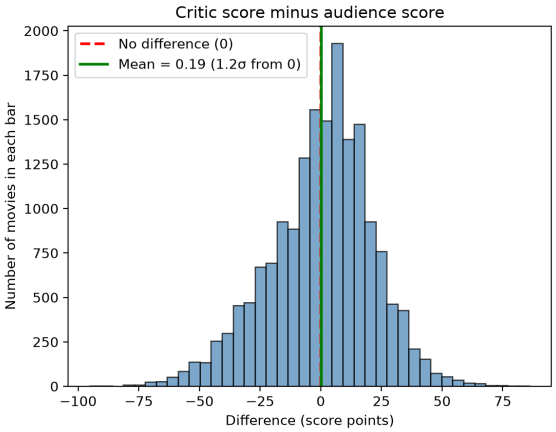


Figure 1: Critic minus audience score for 17,407 movies.

Does this depend on the movie genre? If we look at figure 2, we can see that six genres show a difference larger than 3σ (Figure 2). Critics rate documentaries and classics about 7 points higher than audiences, and art house & international and horror about 4–5 points higher. Audiences rate comedy and action & adventure about 4 points higher. For the other genres the error bars cross zero, so there is no clear difference.

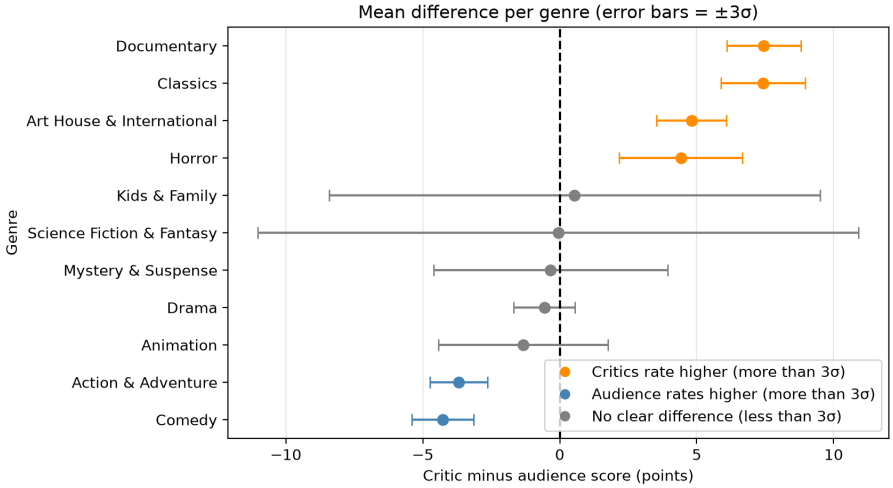


Figure 2: Mean difference per genre with $\pm 3\sigma$ error bars. Orange: critics rate higher; blue: audience rates higher; grey: no clear difference.

4 Conclusion and limitations

On average, critics and audiences agree (1.2σ). But this hides real differences: in six genres they clearly disagree, in both directions. Critics prefer documentaries, classics, art house and horror, while audiences prefer comedy and action. These opposite effects cancel out in the overall average.

5 Limitations.

Only the first listed genre was used for each movie, so movies with several genres are only included in one. 305 movies without scores were left out. Finally, the subsampling shows that the result is stable, but not that the movies are independent of each other.